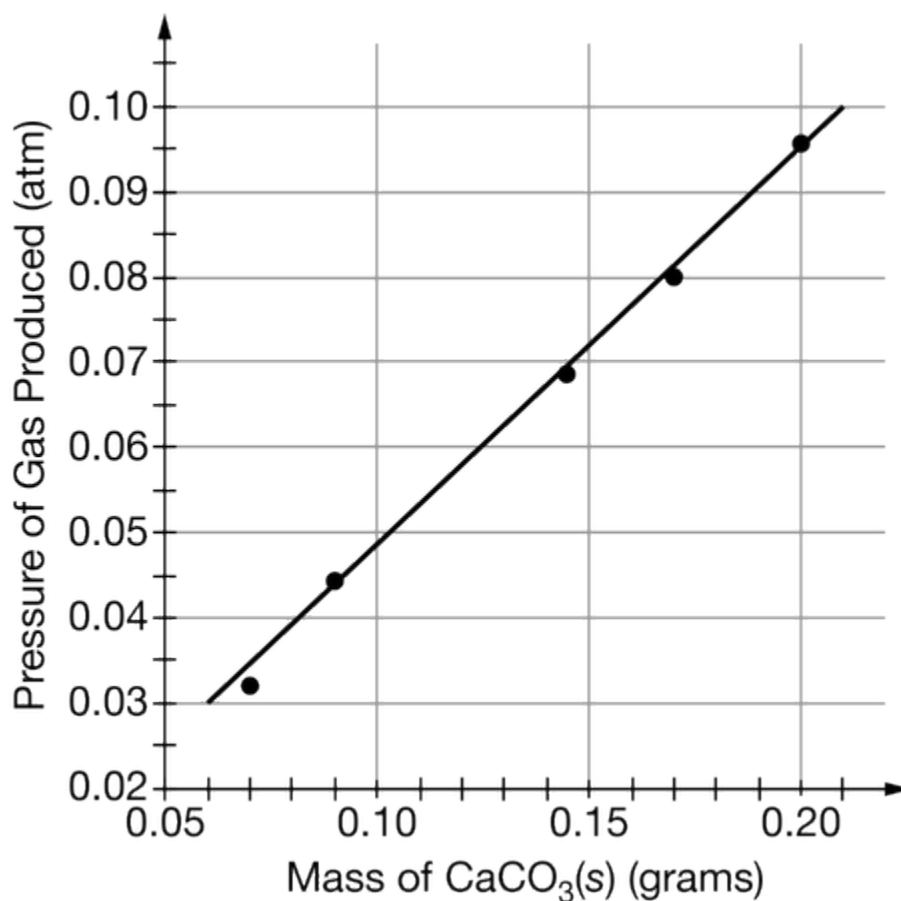


Chapter 04 Part 1: Changes and Representations

An experiment is performed to measure the mass percent of $\text{CaCO}_3(s)$ in eggshells. Five different samples of $\text{CaCO}_3(s)$ of known mass react with an excess of $2.0\text{ M HCl}(aq)$ in identical sealed, rigid reaction vessels. The pressure of the gas produced is measured with a pressure gauge attached to the reaction vessel. Since the reaction is exothermic, the reaction system is cooled to its original temperature before the pressure is recorded. The experimental data are used to create the calibration line below.



The experiment is repeated with an eggshell sample, and the experimental data are recorded in the table below.

Mass of eggshell sample	0.200 g
Pressure prior to reaction	0.800 atm
Pressure at completion of reaction	0.870 atm

- Which of the following equations best represents the species that react and the species that are produced when $\text{CaCO}_3(s)$ and $\text{HCl}(aq)$ are combined?

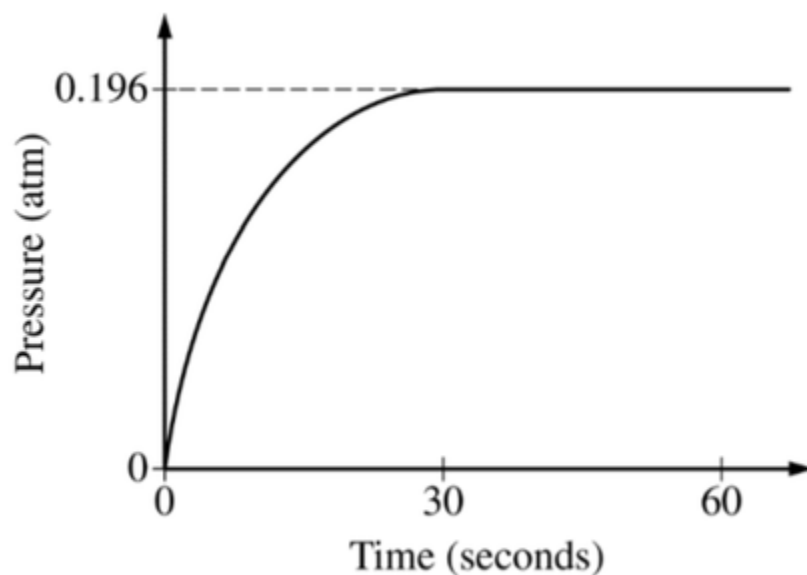
Chapter 04 Part 1: Changes and Representations

- (A) $2\text{H}^+(aq) + \text{CaCO}_3(s) \rightarrow \text{Ca}^{2+}(aq) + \text{CO}_3^{2-}(aq) + \text{H}_2(g)$
- (B) $2\text{H}^+(aq) + \text{CaCO}_3(s) \rightarrow \text{Ca}^{2+}(aq) + \text{H}_2\text{O}(l) + \text{CO}_2(g)$
- (C) $2\text{H}^+(aq) + \text{CaCO}_3(s) \rightarrow \text{Ca}^{2+}(aq) + 2\text{OH}^-(aq) + \text{CO}(g)$
- (D) $2\text{HCl}(aq) + \text{CaCO}_3(s) \rightarrow \text{Ca}^{2+}(aq) + \text{H}_2\text{O}(l) + \text{CO}_2(g) + 2\text{Cl}^-(aq)$
-

Chapter 04 Part 1: Changes and Representations

Substance	Lewis Diagram	Boiling Point
CH_3OH	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\ddot{\text{O}}-\text{H} \\ \\ \text{H} \end{array}$	338 K
$\text{C}_2\text{H}_5\text{OH}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\ddot{\text{O}}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	351 K

Equimolar samples of $\text{CH}_3\text{OH}(l)$ and $\text{C}_2\text{H}_5\text{OH}(l)$ are placed in separate, previously evacuated, rigid 2.0 L vessels. Each vessel is attached to a pressure gauge, and the temperatures are kept at 300 K. In both vessels, liquid is observed to remain present at the bottom of the container at all times. The change in pressure inside the vessel containing $\text{CH}_3\text{OH}(l)$ is shown below.



2. Which of the following best describes the change that takes place immediately after the $\text{CH}_3\text{OH}(l)$ is introduced into the previously evacuated vessel?

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- (A) A chemical change takes place because covalent bonds are broken.
 - (B) A chemical change takes place because intermolecular attractions are overcome.
 - (C) A physical change takes place because covalent bonds are broken.
 - (D) A physical change takes place because intermolecular attractions are overcome.
-

3. Which of the following describes the changes in forces of attraction that occur as H_2O changes phase from a liquid to a vapor?

- (A) $\text{H}-\text{O}$ bonds break as $\text{H}-\text{H}$ and $\text{O}-\text{O}$ bonds form.
- (B) Hydrogen bonds between H_2O molecules are broken.
- (C) Covalent bonds between H_2O molecules are broken.
- (D) Ionic bonds between H^+ ions and OH^- ions are broken.
- (E) Covalent bonds between H^+ ions and H_2O molecules become more effective.

4.

Mass of empty crucible	12.0 g
Mass of crucible and sample before heating	14.4 g
Mass of crucible and sample after heating	16.0 g

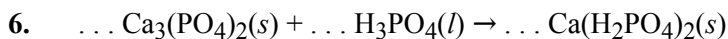
A student places a sample of a pure metal in a crucible and heats it strongly in air. Data from the experiment are given in the table above. The final mass was determined after the sample was cooled to room temperature. Which of the following statements related to the experiment is correct?

- (A) The mass of the sample decreased, so physical changes occurred as the metal first melted and then boiled out of the crucible.
- (B) The mass of the sample increased, so a chemical change occurred when bonds formed between the metal and another substance.
- (C) There was nothing for the metal to react with, so only a physical change could have occurred.
- (D) The sample was only heated, so neither a physical nor a chemical change occurred.

5. $\dots \text{C}_{10}\text{H}_{12}\text{O}_4\text{S}(s) + \dots \text{O}_2(g) \rightarrow \dots \text{CO}_2(g) + \dots \text{SO}_2(g) + \dots \text{H}_2\text{O}(g)$

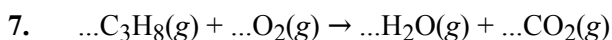
When the equation above is balanced and all coefficients are reduced to their lowest whole-number terms, the coefficient for $\text{O}_2(g)$ is

- (A) 6
- (B) 7
- (C) 12
- (D) 14
- (E) 28

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When the equation above is balanced and all coefficients are reduced to lowest whole-number terms, what is the coefficient for $\text{H}_3\text{PO}_4(l)$?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5



When the equation for the reaction represented above is balanced and all coefficients are reduced to the lowest whole-number terms, the coefficient for $\text{O}_2(g)$ is

- (A) 1
- (B) 2
- (C) 3
- (D) 5
- (E) 6

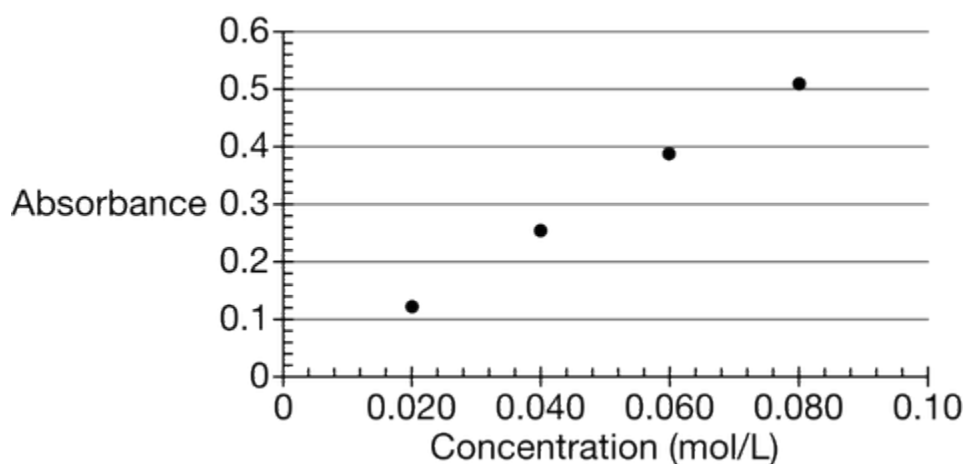
8. A student mixes 20.0 g of white KCl crystals with distilled water in a beaker. After the mixture was stirred, no crystals are visible and the solution is clear. After several days, all of the water evaporates and white crystals are found in the beaker. Which of following pieces of experimental evidence would best help the student to confirm that a new compound had not been made and that only a physical change occurred?

- (A) The solution does not change color after stirring.
- (B) The KCl crystals are no longer visible after mixing with water.
- (C) There is a temperature change in the solution during the dissolving process.
- (D) After the water has evaporated, the white crystals in the beaker have a mass of 20.0 g.

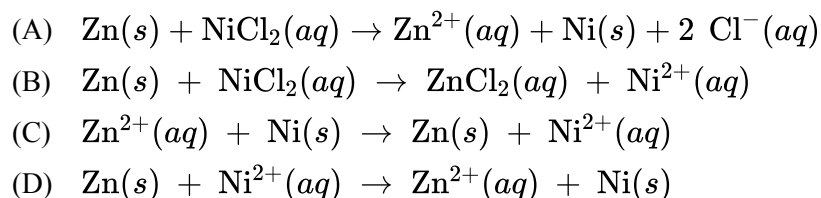
Chapter 04 Part 1: Changes and Representations

An experiment was performed to investigate the reaction between Zn metal and $\text{Ni}^{2+}(aq)$ at different concentrations. Because the $\text{Ni}^{2+}(aq)$ ion is green, the extent of the reaction was determined using spectrophotometric analysis. Four 20.0 mL standard solutions of $\text{Ni}^{2+}(aq)$ were prepared by dissolving $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ (molar mass 240 g/mol) in water. The absorbance of each solution was measured; the results are shown both in the table below and in the following plot of the absorbance data.

Solution	$[\text{Ni}^{2+}]$	Absorbance
1	0.020	0.12
2	0.040	0.25
3	0.060	0.39
4	0.080	0.51



9. Which of the following equations best represents the reacting species in the reaction between Zn metal and $\text{NiCl}_2(aq)$?



10. $\dots \text{LiHCO}_3(aq) + \dots \text{H}_2\text{SO}_4(aq) \rightarrow \dots \text{Li}_2\text{SO}_4(aq) + \dots \text{H}_2\text{O}(l) + \dots \text{CO}_2(g)$

When the equation above is balanced and the coefficients are reduced to lowest whole-number terms, what is the coefficient of $\text{H}_2\text{O}(l)$?

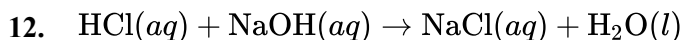
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- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5



The exothermic process represented above is best classified as a

- (A) physical change because a new phase appears in the products
- (B) physical change because $\text{O}_2(g)$ that was dissolved comes out of solution
- (C) chemical change because entropy increases as the process proceeds
- (D) chemical change because covalent bonds are broken and new covalent bonds are formed



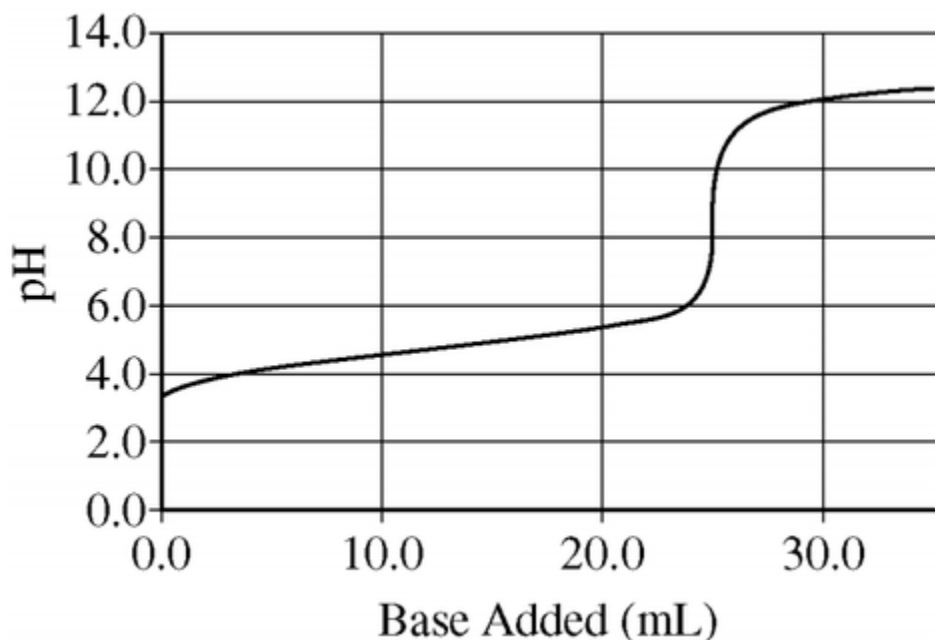
A student had two dilute, colorless solutions, $\text{HCl}(aq)$ and $\text{NaOH}(aq)$, which were at the same temperature. The student combined the solutions, and the reaction represented above occurred. Which of the following results would be evidence that a chemical reaction took place?

- (A) The resulting solution is colorless.
- (B) The temperature of the reaction mixture increases.
- (C) The total volume of the mixture is approximately equal to the sum of the initial volumes.
- (D) The resulting solution conducts electricity.

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A 0.35 g sample of $\text{Li}(s)$ is placed in an Erlenmeyer flask containing 100 mL of water at 25°C . A balloon is placed over the mouth of the flask to collect the hydrogen gas that is generated.

After all of the $\text{Li}(s)$ has reacted with $\text{H}_2\text{O}(l)$, the solution in the flask is added to a clean, dry buret and used to titrate an aqueous solution of a monoprotic acid. The pH curve for this titration is shown in the diagram below



13. Which of the following is the balanced net-ionic equation for the reaction between $\text{Li}(s)$ and water?
- (A) $2 \text{Li}(s) + 2 \text{H}^+(aq) + 2 \text{OH}^-(aq) \rightarrow 2 \text{Li}^+(aq) + 2 \text{OH}^-(aq) + \text{H}_2(g)$
- (B) $2 \text{Li}(s) + 2 \text{H}_2\text{O}(l) \rightarrow 2 \text{Li}^+(aq) + 2 \text{OH}^-(aq) + \text{H}_2(g)$
- (C) $2 \text{Li}(s) + 2 \text{H}_2\text{O}(l) \rightarrow 2 \text{LiOH}(s) + \text{H}_2(g)$
- (D) $2 \text{Li}(s) + 2 \text{H}_2\text{O}(l) \rightarrow 2 \text{LiH}(s) + \text{H}_2(g)$
-
14. A student was asked to formulate a hypothesis about what would happen if 100. mL of 0.1 M $\text{NaOH}(aq)$ at 25°C was combined with 100. mL of 0.1 M $\text{MgCl}_2(aq)$ at 25°C . Which of the following hypotheses indicates that the student thought a chemical change would occur?
- (A) The volume of the resulting solution will be equal to the sum of the volumes of the original solutions.
- (B) The mass of the resulting solution will be equal to the sum of the masses of the original solutions.
- (C) The resulting solution would contain a precipitate.
- (D) The resulting solution will be clear.

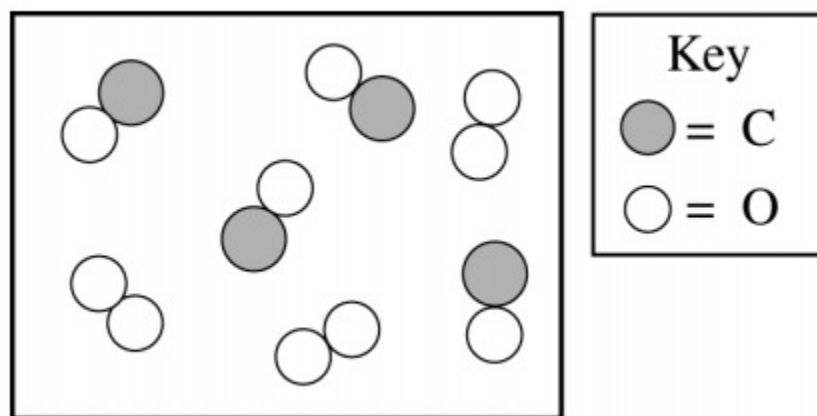
A 0.50 g sample of $\text{Mg}(s)$ was placed in a solution of $\text{HCl}(aq)$, where it reacted completely.

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15. Which of the following equations best represents the reacting species in the reaction described above?
- (A) $\text{Mg}(s) + 2 \text{H}^+(aq) \rightarrow \text{Mg}^{2+}(aq) + \text{H}_2(g)$
- (B) $\text{Mg}(s) + \text{HCl}(aq) \rightarrow \text{MgCl}_2(aq) + \text{H}_2(g)$
- (C) $\text{Mg}(s) + 2 \text{HCl}(aq) \rightarrow \text{MgCl}_2(s) + 2 \text{H}^+(aq)$
- (D) $\text{Mg}(s) + 2 \text{HCl}(aq) \rightarrow \text{Mg}^{2+}(aq) + \text{Cl}_2(g) + \text{H}_2(g)$
-
16. Which of the following is the net ionic equation for the reaction between aqueous sodium fluoride and hydrochloric acid?
- (A) $\text{NaF}(aq) + \text{HCl}(aq) \rightarrow \text{NaCl}(aq) + \text{HF}(aq)$
- (B) $\text{Na}^+(aq) + \text{F}^-(aq) + \text{H}^+(aq) + \text{Cl}^-(aq) \rightarrow \text{Na}^+(aq) + \text{Cl}^-(aq) + \text{HF}(aq)$
- (C) $\text{Na}^+(aq) + \text{Cl}^-(aq) \rightarrow \text{NaCl}(aq)$
- (D) $\text{F}^-(aq) + \text{H}^+(aq) \rightarrow \text{HF}(aq)$
17. Equal volumes of 0.2 M solutions of lead(II) nitrate and potassium bromide are combined to form lead(II) bromide as a yellow precipitate. Which of the following is the correct net ionic equation for the reaction?
- (A) $\text{Pb}^{2+}(aq) + 2 \text{Br}^-(aq) \rightarrow \text{PbBr}_2(s)$
- (B) $\text{K}^+(aq) + \text{NO}_3^-(aq) \rightarrow \text{KNO}_3(aq)$
- (C) $\text{Pb}^{2+}(aq) + 2 \text{NO}_3^-(aq) + 2 \text{K}^+(aq) + 2 \text{Br}^-(aq) \rightarrow 2 \text{K}^+(aq) + 2 \text{NO}_3^-(aq) + \text{PbBr}_2(s)$
- (D) $\text{Pb}(\text{NO}_3)_2(aq) + 2 \text{KBr}(aq) \rightarrow \text{PbBr}_2(s) + 2 \text{KNO}_3(aq)$
18. The reaction between aqueous strontium chloride (SrCl_2) and aqueous potassium sulfate (K_2SO_4) forms a precipitate of strontium sulfate (SrSO_4). Which of the following represents the net ionic equation for the reaction?
- (A) $\text{Cl}^-(aq) + \text{K}^+(aq) \rightarrow \text{KCl}(aq)$
- (B) $\text{Sr}^{2+}(aq) + \text{SO}_4^{2-}(aq) \rightarrow \text{SrSO}_4(s)$
- (C) $\text{Sr}^{2+}(aq) + 2 \text{Cl}^-(aq) + \text{SO}_4^{2-}(aq) + 2 \text{K}^+(aq) \rightarrow \text{SrSO}_4(s) + 2 \text{Cl}^-(aq) + 2 \text{K}^+(aq)$
- (D) $\text{SrCl}_2(aq) + \text{K}_2\text{SO}_4(aq) \rightarrow \text{SrSO}_4(s) + 2 \text{KCl}(aq)$
19. A 20. mL sample of 0.50 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ is titrated with 0.50 M $\text{NaOH}(aq)$. Which of the following best represents the species that react and the species produced in the reaction?
- (A) $\text{H}^+(aq) + \text{OH}^-(aq) \rightarrow \text{H}_2\text{O}(l)$
- (B) $\text{H}^+(aq) + \text{C}_2\text{H}_3\text{O}_2^-(aq) + \text{Na}^+(aq) + \text{OH}^-(aq) \rightarrow \text{H}_2\text{O}(l) + \text{NaC}_2\text{H}_3\text{O}_2(aq)$
- (C) $\text{HC}_2\text{H}_3\text{O}_2(aq) + \text{OH}^-(aq) \rightarrow \text{C}_2\text{H}_3\text{O}_2^-(aq) + \text{H}_2\text{O}(l)$
- (D) $\text{HC}_2\text{H}_3\text{O}_2(aq) + \text{NaOH}(aq) \rightarrow \text{H}_2\text{O}(l) + \text{Na}^+(aq) + \text{C}_2\text{H}_3\text{O}_2^-(aq)$

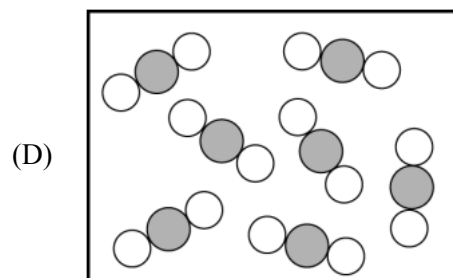
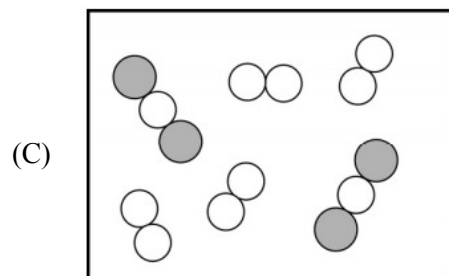
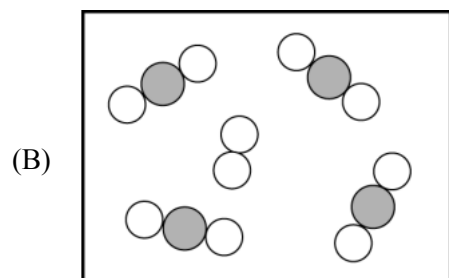
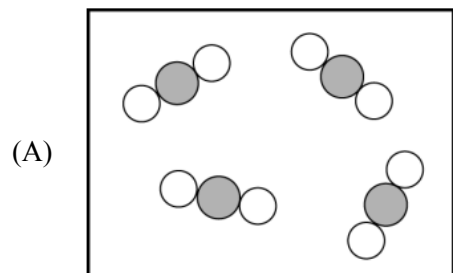
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20.

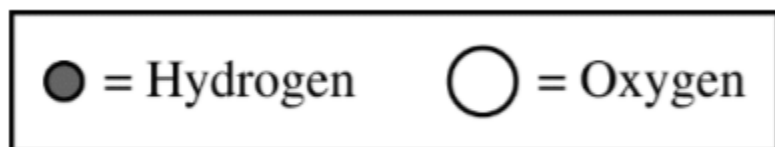


A mixture of $\text{CO}(g)$ and $\text{O}_2(g)$ is placed in a container, as shown above. A reaction occurs, forming $\text{CO}_2(g)$. Which of the following best represents the contents of the box after the reaction has proceeded as completely as possible?

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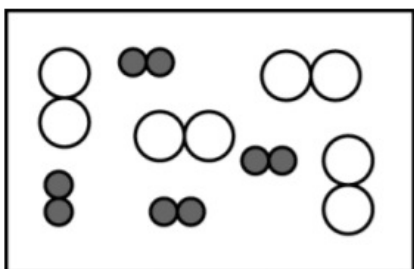


21. Which of the following particle diagrams best represents the products when four molecules of $\text{H}_2\text{O}_2(l)$ decompose into water and oxygen gas at room temperature?

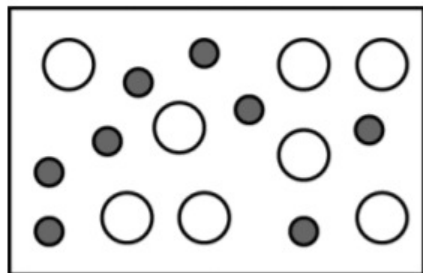


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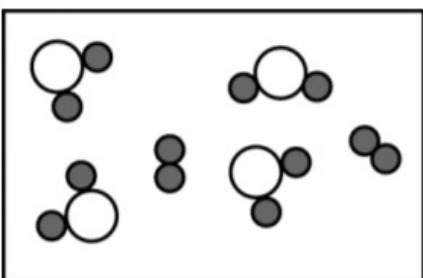
(A)



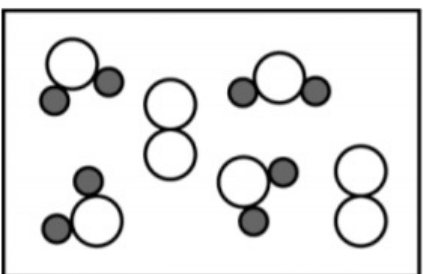
(B)



(C)

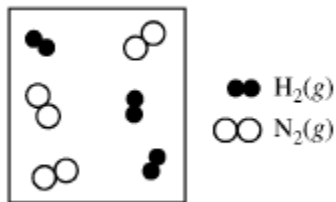


(D)

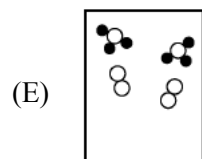
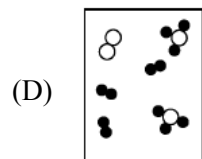
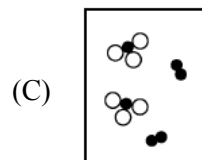
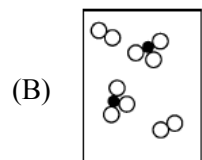
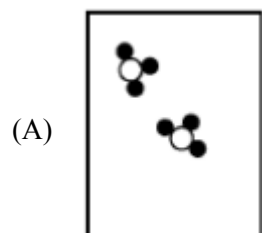
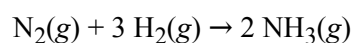


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22.

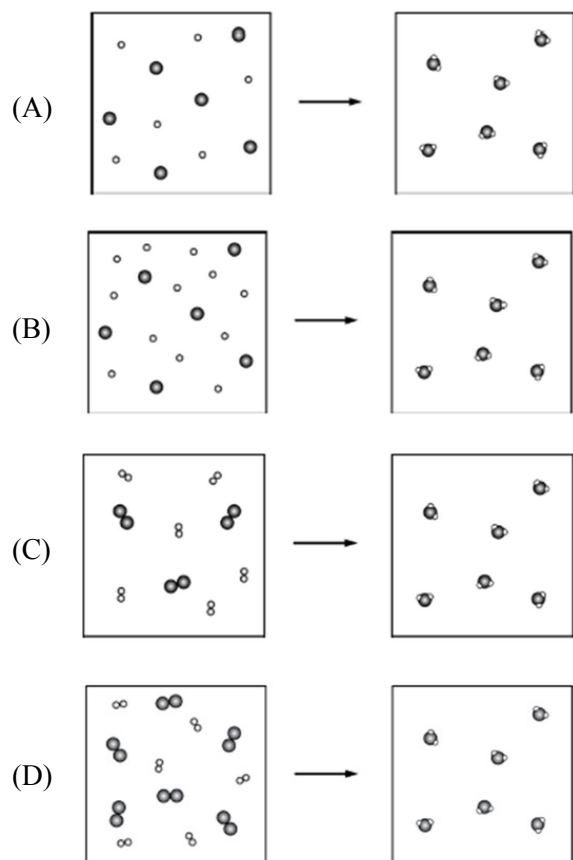


The diagram above represents $\text{H}_2(\text{g})$ and $\text{N}_2(\text{g})$ in a closed container. Which of the following diagrams would represent the results if the reaction shown below were to proceed as far as possible?



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23. Which of the following particulate diagrams best shows the formation of water vapor from hydrogen gas and oxygen gas in a rigid container at 125°C ?



24. A student was studying physical and chemical changes. The student carried out some procedures in the laboratory and recorded observations. For one of the procedures, the student concluded that a physical change took place, but not a chemical change. Which of the following could have been the results of the procedure?
- (A) A cube of metal was changed into a flat sheet of metal.
 - (B) When two liquids at room temperature were combined in a beaker, the beaker became hot.
 - (C) When two clear liquids were combined, the resulting mixture was cloudy.
 - (D) When a colorless liquid was added to a blue liquid, the resulting solution was yellow.

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25.

Substance	Appearance	Boiling Point	Density
Water	Clear, colorless liquid	100.0 °C	1.00 g/mL
X	White, crystalline solid	1935.5 °C	2.15 g/mL
Mixture of water + X	Clear, colorless liquid	101.6 °C	1.08 g/mL

The table above summarizes data given to a student to evaluate the type of change that took place when substance X was mixed with water. The student claimed that the data did not provide enough evidence to determine whether a chemical or physical change took place and that additional tests were needed. Which of the following identifies the best way to gather evidence to support the type of change that occurred when water and X were mixed?

- (A) Measuring the melting point of the mixture of water and X
- (B) Adding another substance to the mixture of water and X to see whether a solid forms
- (C) Measuring and comparing the masses of the water, X, and the mixture of water and X
- (D) Measuring the electrical conductivities of X and the mixture of water and X

26. A student placed a piece of $\text{Zn}(s)$ in a beaker containing $\text{CuSO}_4(aq)$. After 45 minutes, the student observed a brown solid at the bottom of the beaker and that the solution had turned from blue to colorless. Which of the following balanced equations best represents only the species that react and the species that are produced in the reaction?

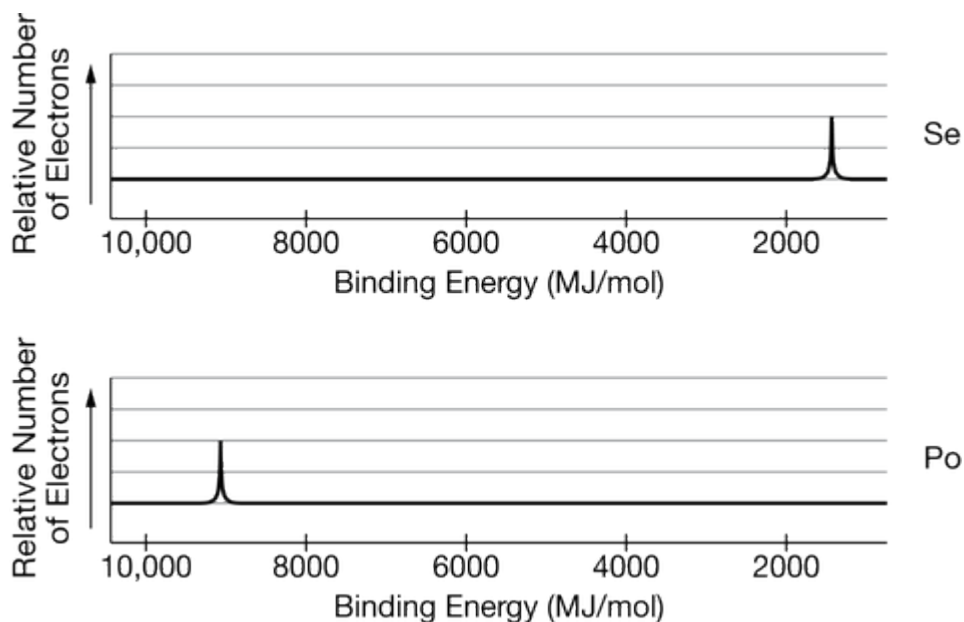
- (A) $2 \text{Zn}(s) + \text{CuSO}_4(aq) \rightarrow \text{Zn}_2\text{SO}_4(aq) + \text{Cu}(s)$
- (B) $\text{Zn}(s) + \text{CuSO}_4(aq) \rightarrow \text{ZnSO}_4(aq) + \text{Cu}(s)$
- (C) $\text{Zn}(s) + \text{Cu}^{2+}(aq) \rightarrow \text{Zn}^{2+}(aq) + \text{Cu}(s)$
- (D) $\text{Zn}(s) + \text{Cu}^+(aq) \rightarrow \text{Zn}^+(aq) + \text{Cu}(s)$

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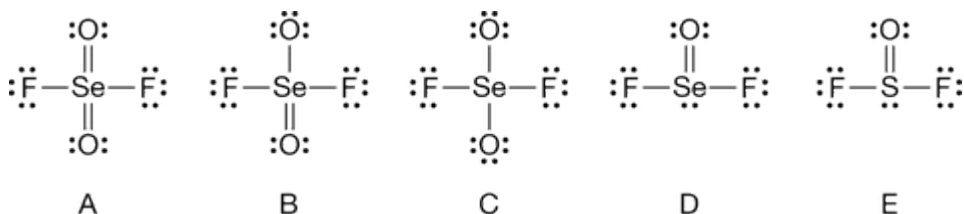
27. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about elements in Group 16 of the periodic table.

- (a) Using principles of atomic structure, explain why the first ionization energy of Po is less than that of Se.
- (b) Photoelectron spectroscopy data for the $1s$ sublevel of Se and the $1s$ sublevel of Po are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Po is positioned so far to the left of the peak for Se.



Three compounds that contain elements from Group 16 are SeO_2F_2 , SeOF_2 , and SOF_2 . Possible Lewis electron dot diagrams for the compounds are shown below.



- (c) Which structure, A, B, or C, minimizes the formal charges for each atom in SeO_2F_2 ? Justify your choice.
- (d) A student proposes that the energy of the $\text{Se} - \text{to} - \text{O}$ bond in the structure in diagram D is greater than that of the $\text{Se} - \text{to} - \text{O}$ bond in the structure in diagram C. Do you agree or disagree with the student's claim? Justify your answer.

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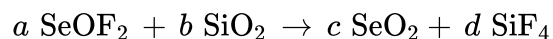
(e) The dipole moment of the SeO_2F_2 molecule (diagrams A, B and C) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The melting point of SeOF_2 (diagram D) is greater than the melting point of SOF_2 (diagram E).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) SeOF_2 reacts with SiO_2 to produce SeO_2 and SiF_4 . The unbalanced chemical equation for the reaction appears below.



Give the value of each of the following coefficients in the balanced equation.

$a =$

$b =$

$c =$

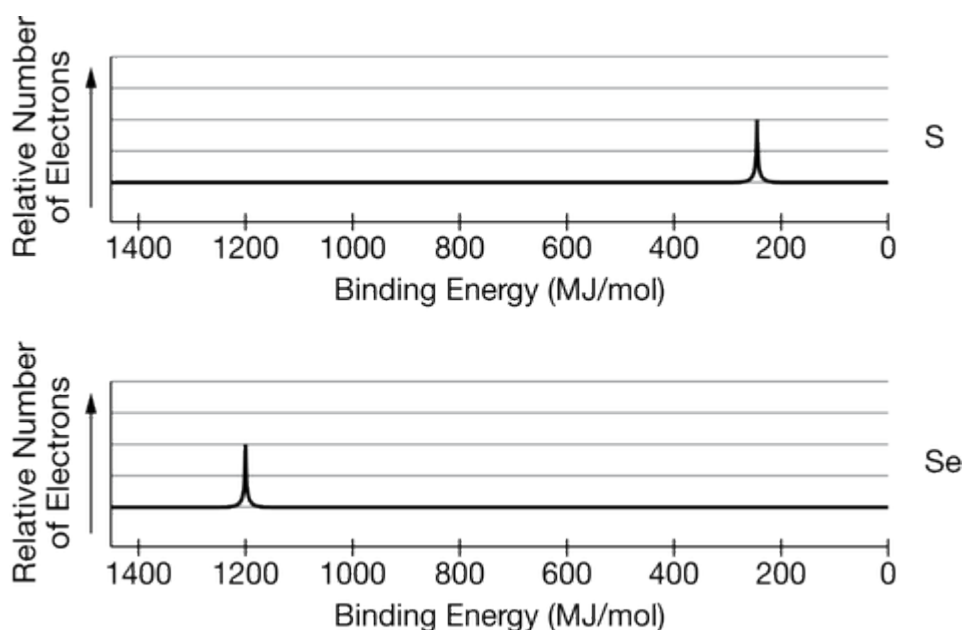
$d =$

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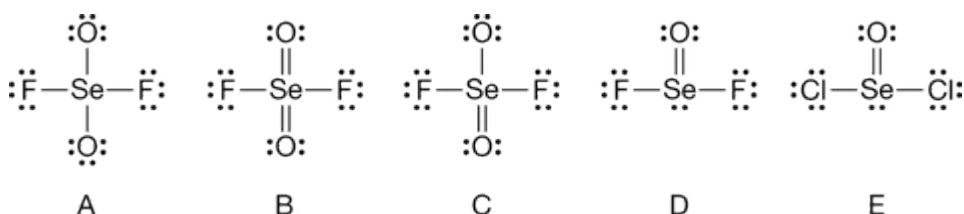
28. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about elements in Group 16 of the periodic table.

- (a) Using principles of atomic structure, explain why the atomic radius of Kr is less than the ionic radius of Se^{2-} .
- (b) Photoelectron spectroscopy data for the $1s$ sublevel of S and the $1s$ sublevel of Se are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Se is positioned so far to the left of the peak for S.



Three compounds that contain elements from Group 16 are SeO_2F_2 , SeOF_2 , and SeOCl_2 . Possible Lewis electron dot diagrams for the compounds are shown below.



- (c) Which structure, A, B, or C, minimizes the formal charges for each atom in SeO_2F_2 ? Justify your choice.
- (d) A student proposes that the energy of the $\text{Se} - \text{to} - \text{O}$ bond in the structure in diagram D is less than that of

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the Se — to — O bond in the structure in diagram A. Do you agree or disagree with the student's claim? Justify your answer.

(e) The dipole moment of the SeOF_2 molecule (diagram D) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The boiling point of SeOF_2 (diagram D) is less than the boiling point of SeOCl_2 (diagram E).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in boiling points based on the types and relative strengths of intermolecular forces.

(h) SeOF_2 reacts with NaOH to produce Na_2SeO_3 , NaF , and H_2O . The unbalanced chemical equation for the reaction appears below.



Give the value of each of the following coefficients in the balanced equation.

$a =$

$b =$

$c =$

$d =$

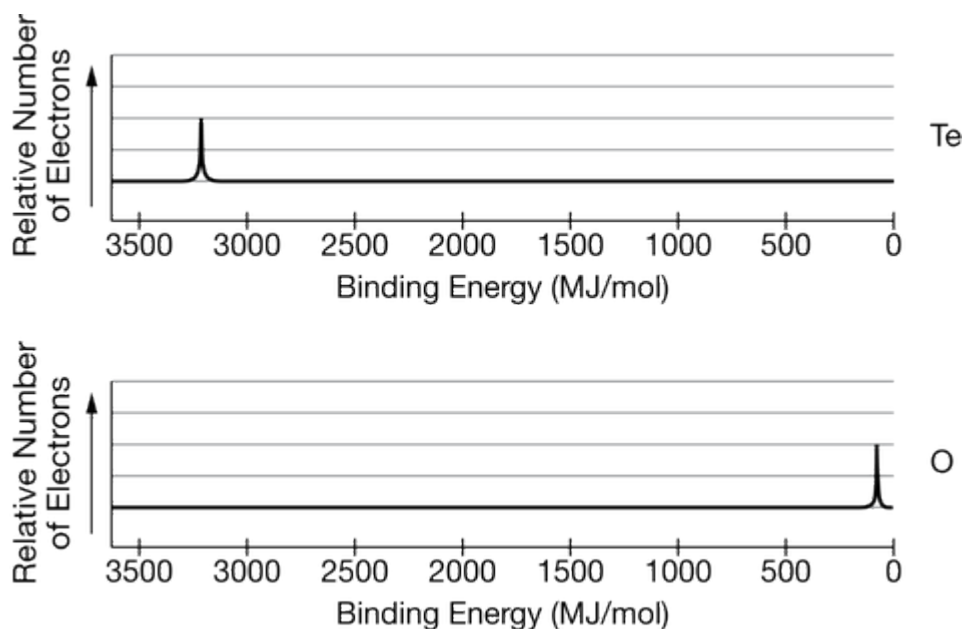
$e =$

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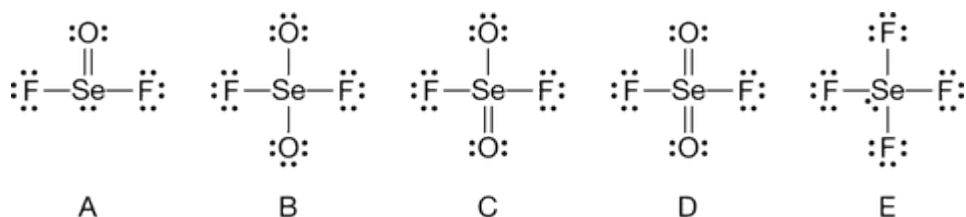
29. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about elements in Group 16 of the periodic table.

- (a) Using principles of atomic structure, explain why the atomic radius of Te is less than the ionic radius of Te^{2-} .
- (b) Photoelectron spectroscopy data for the $1s$ sublevel of Te and the $1s$ sublevel of O are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for O is positioned so far to the right of the peak for Te.



Three compounds that contain elements from Group 16 are SeOF_2 , SeO_2F_2 , and SeF_4 . Possible Lewis electron dot diagrams for the compounds are shown below.



- (c) Which structure, B, C or D, minimizes the formal charges for each atom in SeO_2F_2 ? Justify your choice.
- (d) A student proposes that the energy of the $\text{Se}-\text{to}-\text{F}$ bond in the structure in diagram A is equal to that of the $\text{Se}-\text{to}-\text{F}$ bond in the structure in diagram D. Do you agree or disagree with the student's claim? Justify

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your answer.

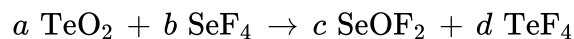
(e) The dipole moment of the SeF_4 molecule (diagram E) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The melting point of SeOF_2 (diagram A) is greater than the melting point of SeO_2F_2 (diagrams B, C, and D).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) TeO_2 reacts with SeF_4 to produce SeOF_2 and TeF_4 . The unbalanced chemical equation for the reaction appears below.



Give the value of each of the following coefficients in the balanced equation.

$a =$

$b =$

$c =$

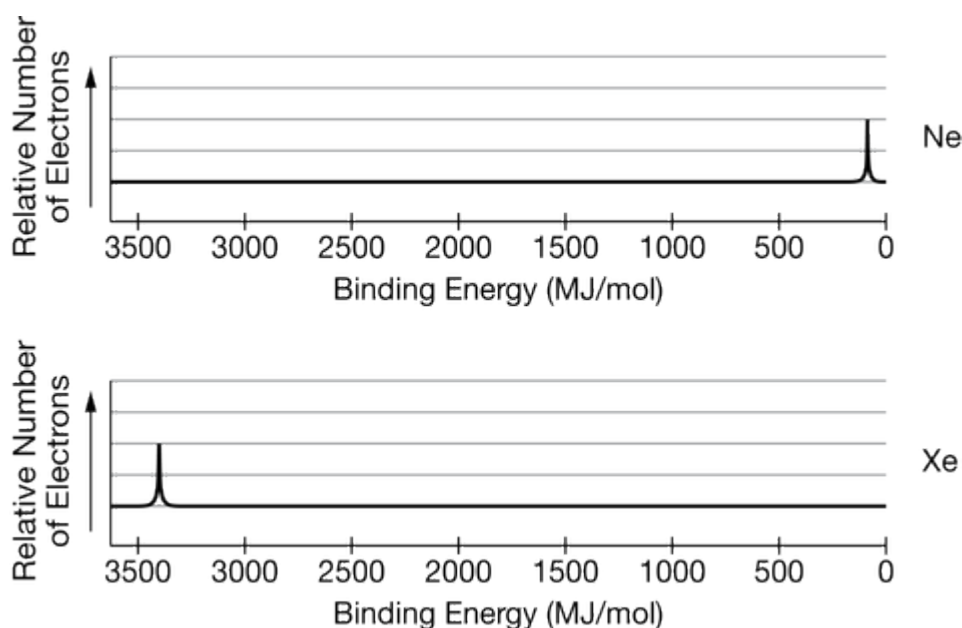
$d =$

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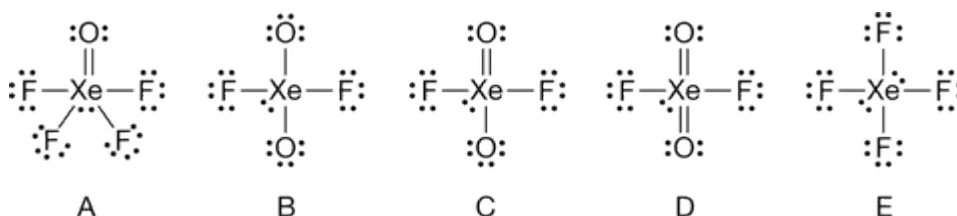
30. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about elements in Group 18 of the periodic table.

- (a) Using principles of atomic structure, explain why the atomic radius of Xe is less than that of Te.
- (b) Photoelectron spectroscopy data for the $1s$ sublevel of Ne and the $1s$ sublevel of Xe are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Ne is positioned so far to the right of the peak for Xe.



Three compounds that contain an element from Group 18 are XeOF_4 , XeO_2F_2 , and XeF_4 . Possible Lewis electron dot diagrams for the compounds are shown below.



- (c) Which structure, B, C, or D, minimizes the formal charges for each atom in XeO_2F_2 ? Justify your choice.
- (d) A student proposes that the energy of the $\text{Xe} - \text{to} - \text{F}$ bond in the structure in diagram B is equal to that of the $\text{Xe} - \text{to} - \text{F}$ bond in the structure in diagram A. Do you agree or disagree with the student's claim? Justify your answer.

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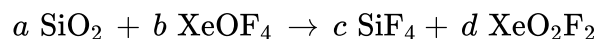
(e) The dipole moment of the XeOF_4 molecule (diagram A) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The boiling point of XeF_4 (diagram E) is greater than the boiling point of elemental Xe.

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in boiling points based on the types and relative strengths of intermolecular forces.

(h) XeOF_4 reacts with SiO_2 to produce XeO_2F_2 and SiF_4 . The unbalanced chemical equation for the reaction appears below.



Give the value of each of the following coefficients in the balanced equation.

$a =$

$b =$

$c =$

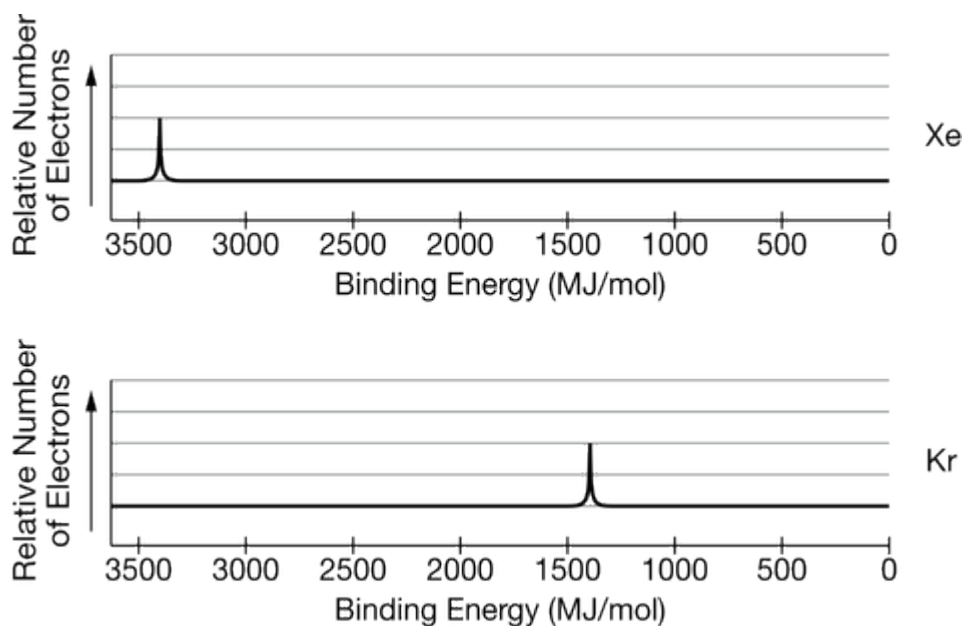
$d =$

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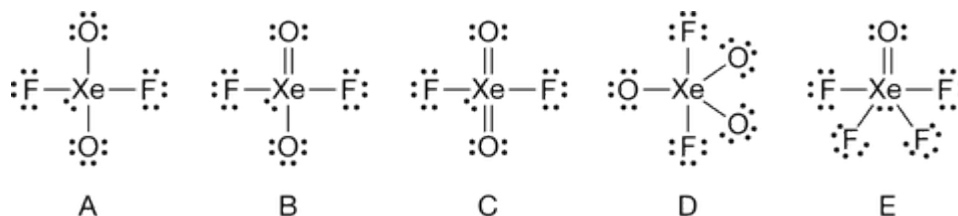
31. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about elements in Group 18 of the periodic table.

- (a) Using principles of atomic structure, explain why the ionic radius of Cs^+ is less than the atomic radius of Xe.
- (b) Photoelectron spectroscopy data for the $1s$ sublevel of Xe and the $1s$ sublevel of Kr are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Kr is positioned so far to the right of the peak for Xe.



Three compounds that contain an element from Group 18 are XeO_2F_2 , XeO_3F_2 , and XeOF_4 . Possible Lewis electron dot diagrams for the compounds are shown below.



- (c) Which structure, A, B, or C, minimizes the formal charges for each atom in XeO_2F_2 ? Justify your choice.
- (d) A student proposes that the energy of the Xe — to — O bond in the structure in diagram E is greater than that of the Xe — to — O bond in the structure in diagram D. Do you agree or disagree with the student's claim? Justify your answer.

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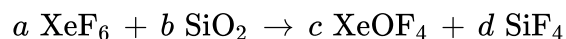
(e) The dipole moment of the XeO_3F_2 molecule (diagram D) is zero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a zero dipole moment.

The melting point of XeOF_4 (diagram E) is greater than the melting point of XeO_3F_2 (diagram D).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) XeF_6 reacts with SiO_2 to produce XeOF_4 and SiF_4 . The unbalanced chemical equation for the reaction appears below.



Give the value of each of the following coefficients in the balanced equation.

$a =$

$b =$

$c =$

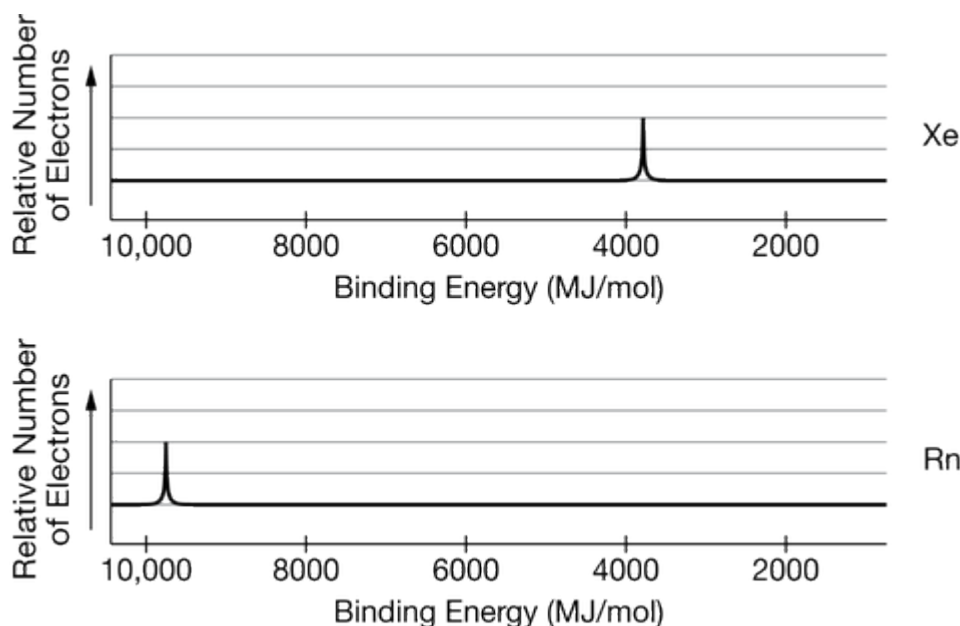
$d =$

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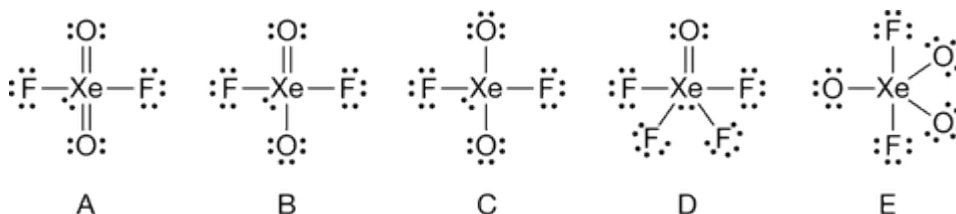
32. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about elements in Group 18 of the periodic table.

- (a) Using principles of atomic structure, explain why the first ionization energy of Rn is less than that of Xe.
- (b) Photoelectron spectroscopy data for the $1s$ sublevel of Xe and the $1s$ sublevel of Rn are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Xe is positioned so far to the right of the peak for Rn.



Three compounds that contain an element from Group 18 are XeO_2F_2 , XeOF_4 , and XeO_3F_2 . Possible Lewis electron-dot diagrams for the compounds are shown below.



- (c) Which structure, A, B or C, minimizes the formal charges for each atom in XeO_2F_2 ? Justify your choice.
- (d) A student proposes that the energy of the Xe — to — O bond in the structure in diagram D is less than that of the Xe — to — O bond in the structure in diagram E. Do you agree or disagree with the student's claim? Justify your answer.

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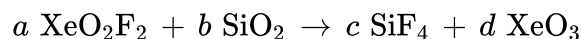
(e) The dipole moment of the XeO_2F_2 molecule (diagrams A, B and C) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The melting point of XeO_2F_2 (diagrams A, B, C) is greater than the melting point of XeO_3F_2 (diagram E).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) XeO_2F_2 reacts with SiO_2 to produce SiF_4 and XeO_3 . The unbalanced chemical equation for the reaction appears below.



Give the value of each of the following coefficients in the balanced equation.

$a =$

$b =$

$c =$

$d =$